

Data Structures in Python

1. Lists in Python

Lists are used to store multiple items in a single variable. Lists are ordered, changeable (mutable), and allow duplicate values.

- **Methods:** `append()`, `insert()`, `remove()`, `pop()`, `clear()`, `sort()`, `reverse()`.
- **Slicing:** You can specify a range of indexes by specifying where to start and where to end the range. `list[start:end:step]`
- **List Comprehension:** Offers a shorter syntax when you want to create a new list based on the values of an existing list. Example: `[x for x in fruits if "a" in x]`

2. Tuples

Tuples are used to store multiple items in a single variable. A tuple is a collection which is ordered and **unchangeable** (immutable).

Tuples are written with round brackets `()`. Because they are immutable, operations like `append()` or `remove()` are not available.

3. Sets

A set is a collection which is unordered, unchangeable (but you can remove items and add new items), and unindexed. Sets do not allow duplicate values.

Sets are written with curly brackets {}. They are highly efficient for membership testing (checking if an item exists) and removing duplicates from a sequence.

Common operations include `union()`, `intersection()`, `difference()`, `symmetric_difference()`.

4. Dictionaries

Dictionaries are used to store data values in key:value pairs. A dictionary is a collection which is ordered (as of Python 3.7), changeable and do not allow duplicates (keys must be unique).

- **Methods:** `get()`, `keys()`, `values()`, `items()`, `update()`, `pop()`.
- **Iteration:** You can loop through a dictionary to get keys, values, or both (using `.items()`).

5. Nested Data Structures

A dictionary can contain dictionaries, this is called nested dictionaries. Similarly, a list can contain lists (nested lists/2D arrays), or a list can contain dictionaries, etc.

Nested structures are very common for representing complex data structures like JSON.